Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years of experience building developer-oriented AI products across APIs, web apps, and async workflows using Python, TypeScript/JavaScript, SQL, and cloud-native tooling.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built core backend product features for a multi-tenant AI platform using **FastAPI**, Azure SQL, and PostgreSQL, supporting 500+ users and reducing dashboard latency from 1.2s to 350ms.
- Designed asynchronous validation/release workflows for 5K+ monthly jobs with reconciliation checks, improving reliability and production readiness for model-driven features.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed product features across a Flask + React stack handling **10K+ daily requests**, including modular APIs, secure ingestion, and user-facing AI workflows.
- Implemented concurrent execution paths and CI/CD controls with GitHub Actions, reducing turnaround time by 60% while maintaining stable API behavior under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built Python experimentation and simulation pipelines for autonomous systems, processing **100K+ trajectory logs** with reproducible runs and measurable performance gains.
- Documented model behavior and debugging findings to support iterative productization and robust handoff across engineering/research teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered API and ETL services with OOP, AWS Glue, and S3 in a microservice architecture; improved response time by 30% and pipeline efficiency by 68%.
- Maintained 20+ unit/integration suites via Jenkins, PyTest, and Mockito across 3 OS environments, accelerating release quality and regression confidence.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built a MERN product surface with REST APIs and WebSockets supporting 8+ workflows and 2M+ synchronized records across user-facing modules.
- Automated deployment and verification with Jenkins and Docker, reducing deployment time by 50% and improving production release consistency.

Skills

Languages: Python, JavaScript/TypeScript, Java, C++, SQL
Core Product Stack: FastAPI, Flask, Node.js, Express.js, React.js, REST APIs, JSON
Data & Infra: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, AWS (S3, RDS, Glue)
Platform Tooling: Docker, Kubernetes, GitHub Actions, Jenkins, Git, CI/CD
AI/ML: LLM workflow integration, asynchronous model pipelines, prompt-driven systems
Engineering: API design, debugging, production support, Agile/Scrum, cross-functional collaboration

Projects

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built semantic retrieval and summarization workflows over **50K+ documents**, enabling sub-second contextual Q&A and production-style AI data access patterns.

Chat2SQL | *Java, TypeScript, SQL, Maven, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases with schema exploration, structured query validation, and reliable execution across Java + TypeScript services.

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built secure product workflows for 50K+ users and 1M+ transactions with real-time updates, scalable deployment patterns, and developer-oriented integration APIs.